VR-Numbers.

Operational Superstructures of Integers, Rationals, Reals, and Complex Numbers over the Natural Numbers of VR

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Abstract

Building on VR (Reznik, 2026, DOI: 10.5281/zenodo.20212092), a formal axiomatization of arithmetic with three primitives {∅, →, t} and four axioms, we develop VR-Numbers — a program of operational superstructures yielding ℤ, ℚ, ℝ, and ℂ over VR's natural numbers ℕ. The construction uses no ordered pairs as ontological objects; each numerical extension is defined as a formal language of expressions with equivalence relations and operations. The ontological foundation of the entire system is the single primitive ∅; all numerical objects — including ℕ — arise as operational constructions of progressively greater depth. The classical actual infinity is replaced by operational infinity: A4 (induction) is read as an operational principle, not as a postulate of completed infinite totalities. The two-dimensionality of ℂ is derived from the duality present in axiom A1 of VR (the two generating implications F → F and F → ⊤), rather than postulated through pairs of reals. ℤ_VR, ℚ_VR, and ℂ_VR are shown to be isomorphic to the corresponding standard structures; ℝ_VR is isomorphic to the field of computable real numbers (a countable subfield of classical ℝ), reflecting the operational character of the construction.

Keywords: foundations of mathematics, operational extensions, operational infinity, integer construction, rational construction, real numbers, complex numbers, Cauchy sequences, axiomatic duality, ontology of numbers.

Part I. Foundation: VR and ℕ

This work builds upon the formal system VR introduced in [Reznik 2026]. We briefly recall its essentials.

Primitives of VR.

Constant: ∅. Binary operator: → (implication). Unary operator: t (succession).

Axioms.

A1 (Generativity). F → F = ⊤ and F → ⊤ = ⊤, where F is identified with ∅ at the logical level, and ⊤ := F → F.

A2 (Implication). → is the classical implication on {F, ⊤}.

A3 (Succession). t(x) = x ∪ {x}, so x ∈ t(x) and x ⊂ t(x).

A4 (Induction). The objects O_n = t^n(∅) exhaust all objects generated from ∅ via t.

Natural Numbers.

$$O_0 = \varnothing,\ O_1 = t(\varnothing) = \{\varnothing\},\ O_2 = t(t(\varnothing)) = \{\varnothing, \{\varnothing\}\},\ \ldots$$

The natural number n is identified with the expression O_n. Equality is defined via Leibniz's principle: x = y ⇔ ∀p: p(x) ↔ p(y). VR has been shown to be arithmetically equivalent to Peano arithmetic, and consistent relative to ZF.

Ontological Status.

In VR, the only ontological primitive is ∅. All other objects of the system — including the natural numbers themselves — arise as operational results of applying the succession operator t to ∅. The natural number O_n is not an independent existing thing; it is the result of n applications of t to ∅. The mathematical content of n is its operational history.

This ontological position is the foundation of the present work. It asserts a minimal ontology — only ∅ exists — and treats all other numerical constructions as operations of progressively greater depth performed on this single primitive.

Part II. Integers $\mathbb{Z}$ as an Operational Superstructure

We construct integers as formal expressions over $\mathbb{N}$ with prescribed equivalence and operations.

II.1. The Formal Language of Subtraction

We extend the language of VR by a binary symbol $\ominus$. An expression of the form

$$a \ominus b, \text{ where } a, b \in \mathbb{N}$$

is a syntactic object intended to denote the operation "subtract b from a". This is not an ordered pair, but a formal record of an operation.

II.2. Equivalence Relation

Two expressions are declared equivalent when they would yield the same result, using only the addition operation already defined on $\mathbb{N}$:

$$a \ominus b \approx c \ominus d \iff a + d = b + c.$$

This relation is reflexive, symmetric, and transitive (direct verification using the commutativity and associativity of addition in $\mathbb{N}$, established in VR).

II.3. Operations

Addition.

$$(a \ominus b) \oplus (c \ominus d) := (a + c) \ominus (b + d).$$

Multiplication.

$$(a \ominus b) \otimes (c \ominus d) := (a \times c + b \times d) \ominus (a \times d + b \times c).$$

Subtraction.

$$(a \ominus b) \boxminus (c \ominus d) := (a + d) \ominus (b + c).$$

Additive Inverse.

$$\ominus(a \ominus b) := b \ominus a.$$

All operations are well-defined on equivalence classes.

II.4. Canonical Form

Every expression $a \ominus b$ can be canonized uniquely:

If $a \geq b$, the canonical form is $(a - b) \ominus 0$.

If $a < b$, the canonical form is $0 \ominus (b - a)$.

II.5. Embedding $\mathbb{N} \to \mathbb{Z}_VR$

The natural number $n \in \mathbb{N}$ embeds into $\mathbb{Z}_VR$ as the class of $n \ominus 0$.

II.6. Isomorphism Theorem

Theorem. The set $\mathbb{Z}_VR$ of equivalence classes of expressions $a \ominus b$ with operations $\oplus$, $\otimes$, $\boxminus$ is a commutative ring with unit, isomorphic to the standard ring of integers $\mathbb{Z}$.

Proof sketch. Define $\varphi: \mathbb{Z}_VR \to \mathbb{Z}$ by $\varphi([a \ominus b]) = a - b$. This map is well-defined on classes, respects operations, and is bijective.

Part III. Rationals ℚ as an Operational Superstructure

Rational numbers are constructed analogously to integers, but over ℤ_VR.

III.1. The Formal Language of Division

We introduce a binary symbol ⊘. An expression of the form

$$a \oslash b, \text{ where } a, b \in \mathbb{Z}_VR \text{ and } b \neq 0_\mathbb{Z}$$

is a syntactic object denoting the operation "divide a by b". The restriction $b \neq 0_\mathbb{Z}$ is syntactic.

III.2. Equivalence Relation

$$a \oslash b \approx_\mathbb{Q} c \oslash d \iff a \otimes d \approx b \otimes c,$$

where ⊗ is the multiplication operation in ℤ_VR.

III.3. Operations

Addition.

$$(a \oslash b) \boxplus (c \oslash d) := (a \otimes d \oplus b \otimes c) \oslash (b \otimes d).$$

Multiplication.

$$(a \oslash b) \boxtimes (c \oslash d) := (a \otimes c) \oslash (b \otimes d).$$

Subtraction.

$$(a \oslash b) \boxminus\oslash (c \oslash d) := (a \otimes d \boxminus b \otimes c) \oslash (b \otimes d).$$

Division.

$$(a \oslash b) \oslash\oslash (c \oslash d) := (a \otimes d) \oslash (b \otimes c), \text{ provided } c \neq 0_\mathbb{Z}.$$

III.4. Canonical Form

Every fraction a ⊘ b can be brought to canonical form by making the denominator positive and reducing to lowest terms.

III.5. Embedding ℤ → ℚ_VR

The integer a ∈ ℤ_VR embeds into ℚ_VR as the class of $a \oslash 1_\mathbb{Z}$, where $1_\mathbb{Z} = 1 \ominus 0$ is the multiplicative identity in ℤ_VR.

III.6. Isomorphism Theorem

Theorem. ℚ_VR with operations ⊞, ⊠, ⊟⊘, ⊘⊘ is a field, isomorphic to the standard field of rationals ℚ.

Part IV. Real Numbers $\mathbb{R}$ via Cauchy Sequences

Real numbers require a qualitatively different construction: while $\mathbb{Z}$ and $\mathbb{Q}$ are syntactic constructions over finite expressions, $\mathbb{R}$ requires infinite processes.

IV.1. Functions as Operational Rules

We extend the language of VR by the notion of function $\mathbb{N} \to \mathbb{Q}_VR$. A function $a: \mathbb{N} \to \mathbb{Q}_VR$ is an operational rule: a finite description (algorithm) producing, for each $n \in \mathbb{N}$, a value $a(n) \in \mathbb{Q}_VR$. Functions are not introduced as sets of ordered pairs; they are operational primitives — finite specifications of infinite correspondences.

A consequence of this definition is that each function is fully captured by its finite description. The collection of all such finite descriptions is itself operationally countable: any function in VR-Numbers can be enumerated through its description. This will have implications for the relationship between $\mathbb{R}_VR$ and the classical $\mathbb{R}$, discussed below.

IV.2. Fundamental (Cauchy) Sequences

A function $a: \mathbb{N} \to \mathbb{Q}_VR$ is fundamental (a Cauchy sequence) if:

$$\forall \varepsilon \in \mathbb{Q}_VR,\ \varepsilon > 0,\ \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N: |a(m) \boxminus_{\oslash} a(n)| < \varepsilon.$$

IV.3. Equivalence of Sequences

$$a \approx_C b \iff \forall \varepsilon > 0,\ \exists N: \forall n \geq N: |a(n) \boxminus_{\oslash} b(n)| < \varepsilon.$$

IV.4. The Construction $\mathbb{R}_VR$

$\mathbb{R}_VR$ is defined as the set of equivalence classes of fundamental sequences with respect to $\approx_C$.

IV.5. Operations

Operations are defined componentwise on representatives:

$$[a] \oplus_{\mathbb{R}} [b] := [c], \text{ where } c(n) = a(n) \boxplus b(n);$$
$$[a] \otimes_{\mathbb{R}} [b] := [c], \text{ where } c(n) = a(n) \boxtimes b(n);$$
$$[a] \boxminus_{\mathbb{R}} [b] := [c], \text{ where } c(n) = a(n) \boxminus_{\oslash} b(n);$$
$$[a] \oslash_{\mathbb{R}} [b] := [c], \text{ where } c(n) = a(n) \oslash_{\oslash} b(n).$$

IV.6. Embedding $\mathbb{Q} \to \mathbb{R}_VR$

Each rational $q \in \mathbb{Q}_VR$ embeds into $\mathbb{R}_VR$ as the class of the constant sequence $a(n) = q$ for all n.

IV.7. Isomorphism Theorem

Theorem. ℝ_VR is a real closed ordered field. It is isomorphic to the field of computable real numbers — a countable subfield of the classical real line ℝ — and contains all reals that admit a finite algorithmic description.

Remark. Strictly speaking, ℝ_VR is not isomorphic to the entire classical ℝ. Classical ℝ has cardinality of the continuum; ℝ_VR, consisting of operational rules each given by a finite description, is countable. The relationship is therefore: ℝ_VR is isomorphic to the computable reals, which constitute a countable real closed subfield of classical ℝ. All real numbers encountered in practice — algebraic numbers, rationals, π, e, and indeed any real definable by a finite description — are captured by ℝ_VR. Non-computable reals of classical ℝ have no place in the operational ontology of VR-Numbers, as they correspond to no finite algorithmic description. This aligns VR-Numbers with the constructive tradition (Bishop, Martin-Löf) and remains consistent with the philosophical position that mathematical objects are operations, not transcendent entities.

IV.8. Operational Character of ℝ

A real number in VR is, formally, an algorithm — a finite description specifying how to compute arbitrarily close rational approximations. The square root of 2, for instance, is represented by an algorithm (such as Newton's method) that, given any precision, produces a rational approximation within that precision. This view aligns naturally with constructive mathematics.

Part V. Complex Numbers $\mathbb{C}$ via the Duality of A1

The construction of complex numbers introduces the central new conceptual element of this work. In standard mathematics, $\mathbb{C}$ is built as ordered pairs of real numbers (a, b) with a special multiplication rule, and the two-dimensionality of $\mathbb{C}$ is postulated through the choice of pairs. We propose instead that the two-dimensionality of $\mathbb{C}$ is derived from the logical foundation of VR — specifically, from the duality present in axiom A1.

V.1. The Duality of A1

Axiom A1 of VR states:

$$F \to F = \top,$$
$$F \to \top = \top.$$

These are two distinct generating facts: F generates both values via implication. The two implications represent two distinct movements from F:

$F \to F$: a movement returning to F itself (self-reference at the logical level).

$F \to \top$: a movement to the opposite value $\top$.

We propose to interpret these two implications as two independent axes of numerical extension.

V.2. The Two Axes

The axis $F \to F$ is the real axis: along this axis, the constructions $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R}$ developed above unfold.

The axis $F \to \top$ is the imaginary axis: along this axis, new operational expressions of the form $b \cdot i$ live, where i is an operational marker indicating the imaginary axis.

The marker i is not a number nor an object — it is a syntactic indicator of which axis a value belongs to.

V.3. The Joining Rule

To make $\mathbb{C}$ a single algebraic structure rather than two disjoint copies of $\mathbb{R}$, the two axes must be coupled. This coupling is captured by the rule:

$$i \otimes i := \ominus 1.$$

This rule is postulated as an axiom of joining: applying the imaginary marker twice is equivalent to negation along the real axis. Without this rule, the two axes would remain independent; with it, they are intertwined into the algebraic structure of $\mathbb{C}$.

V.4. Complex Numbers as Operational Expressions

A complex number is a formal expression

$a \oplus b \cdot i$, where $a, b \in \mathbb{R}_VR$.

This is not an ordered pair (a, b). It is a syntactic record of two values along two distinct axes.

V.5. Equivalence

$a \oplus b \cdot i \approx c \oplus d \cdot i \iff a = c$ (in $\mathbb{R}_VR$) and $b = d$ (in $\mathbb{R}_VR$).

V.6. Operations

Addition.

$(a \oplus b \cdot i) \oplus_{\mathbb{C}} (c \oplus d \cdot i) := (a \oplus c) \oplus (b \oplus d) \cdot i$.

Multiplication (using $i \otimes i = \ominus 1$).

$(a \oplus b \cdot i) \otimes_{\mathbb{C}} (c \oplus d \cdot i) := (a \otimes c \boxminus b \otimes d) \oplus (a \otimes d \oplus b \otimes c) \cdot i$.

Conjugation.

$\mathrm{conj}(a \oplus b \cdot i) := a \oplus (\ominus b) \cdot i$.

Modulus.

$|a \oplus b \cdot i| := \sqrt{}(a \otimes a \oplus b \otimes b)$.

V.7. Embedding $\mathbb{R} \to \mathbb{C}_VR$

Each real $a \in \mathbb{R}_VR$ embeds into $\mathbb{C}_VR$ as the expression $a \oplus 0 \cdot i$.

V.8. Isomorphism Theorem

Theorem. $\mathbb{C}_VR$ with the operations defined above is a field, isomorphic to the standard field of complex numbers $\mathbb{C}$.

V.9. Discussion: Foundation of Two-Dimensionality

In the standard treatment, the dimensionality of $\mathbb{C}$ is a postulate: $\mathbb{C}$ is defined as $\mathbb{R} \times \mathbb{R}$ with a specific multiplication, and there is no deeper justification for why $\mathbb{C}$ has exactly two dimensions.

In VR-Numbers, the two-dimensionality of $\mathbb{C}$ is derived from the logical foundation of the system. Axiom A1 contains exactly two generating implications, $F \to F$ and $F \to \top$. These two implications correspond to two distinct axes of numerical construction. The dimensionality of $\mathbb{C}$ is therefore not arbitrary; it reflects the dual generative structure already present in the logical layer of VR.

This is a non-trivial conceptual claim. It suggests that the structure of complex numbers is not a postulate but a consequence of how implication operates on falsity. The two generating movements from F — one returning to F itself, one moving to $\top$ — are

precisely the two axes that, when joined by the rule $i^2 = \ominus 1$, yield the field of complex numbers. The Fundamental Theorem of Algebra — that $\mathbb{C}$ is algebraically closed — finds, in this construction, a possible deeper origin: the duality is exhaustive at the level of A1, and there is no further direction to extend without leaving the logical foundation of VR.

Part VI. The Ontological Position of VR-Numbers

VR-Numbers adopts a minimal ontology: the only ontological primitive is the void ∅. All other objects — natural, integer, rational, real, and complex numbers — are operational constructions of progressively greater depth performed over ∅.

VI.1. The Single Ontological Primitive

∅ is the unique ontological primitive of the system. It is given as Leibnizian initial datum — that which is, prior to any operation, division, or distinction. Every other object of VR or VR-Numbers is the result of operations applied to ∅.

The succession operator t is not itself an object but a rule of operation. The natural numbers O_n = t^n(∅) are records of successive applications of t. Their content is fully captured by their operational history: O_3 is "three applications of t to ∅", and this is the totality of what O_3 is.

VI.2. Depths of Operationality

All numerical systems constructed in this work are operational, but they exhibit different depths of operational character:

System	Operational Character
∅	The single ontological primitive — that which is, prior to all operation
ℕ	Direct applications of t to ∅: O_n = t^n(∅)
ℤ	Finite syntactic expressions a ⊖ b over ℕ, on the F → F axis
ℚ	Finite syntactic expressions a ⊘ b over ℤ, on the F → F axis
ℝ	Algorithms producing infinite sequences of rationals, on the F → F axis
ℂ	Two-axis syntax over F → F and F → ⊤, joined by $i^2 = \ominus 1$

VI.3. Operational Infinity

The axiom A4 of VR — the principle of induction — is conventionally read as postulating actual infinity: the completed totality of all natural numbers existing as an object. We reject this reading. Within the ontological position of VR-Numbers, A4 does not assert the existence of any infinite object. It formulates an operational principle: any property holding for ∅ and preserved under application of t holds for every result of iterated

application of t. The quantifier "for all n" in A4 is operational, not ontological — it expresses universal applicability of an operation, not membership in a completed set.

We call this operational infinity, in contrast to the classical notion of actual infinity. Operational infinity is mathematically equivalent to actual infinity: every theorem provable using actual infinity in classical mathematics is provable using operational infinity in VR-Numbers. The two notions agree on consequences. They differ in ontological status: actual infinity asserts that an infinite totality exists as an object; operational infinity asserts only that an infinite operation is available.

This distinction is significant. The chief philosophical objection to actual infinity, raised since Aristotle and reformulated by intuitionists and constructivists, is the rejection of completed infinities as objects. Operational infinity bypasses this objection: it does not postulate that the totality of natural numbers is a thing. It postulates that the procedure of unbounded counting from $\varnothing$ is available to us, and that any property preserved by this procedure applies universally. The resulting mathematics is the same; the ontological commitment is minimal.

Within VR-Numbers, all references to "infinite" objects — the natural numbers $\mathbb{N}$ as a domain, the integers $\mathbb{Z}$, the rationals $\mathbb{Q}$, the reals $\mathbb{R}$ as Cauchy sequences, the complex numbers $\mathbb{C}$ — are to be understood operationally. None of these collections is an object. Each is a domain of applicability for operations defined over $\varnothing$.

Accordingly, the quantifier notations used throughout this work — expressions of the form "a, b $\in \mathbb{N}$" or "$\forall \varepsilon \in \mathbb{Q}_VR$, $\varepsilon > 0$" — are operational. They mean: "for any concrete expression of the indicated form". They do not assert the existence of $\mathbb{N}$ or $\mathbb{Q}_VR$ as completed objects. The set-theoretic notation is retained for convenience and conformity with standard mathematical practice; the ontological content is operational throughout.

VI.4. The Gradient of Operationality

The structure of VR-Numbers is a gradient: each successive numerical system corresponds to a deeper operational construction over $\varnothing$.

Natural numbers are the most direct operational result: a single operator t is applied repeatedly to the primitive $\varnothing$.

Integers and rationals introduce a layer of formal syntax: expressions over natural numbers, with equivalence relations defining when distinct expressions denote the same operational result.

Real numbers add infinite operational depth: each real number is itself a finite description (algorithm), but the operation it specifies is infinite — producing arbitrarily close rational approximations.

Complex numbers add a second axis: the duality of A1 generates two independent operational directions, and $\mathbb{C}$ is the syntactic structure on these two axes joined by the multiplication rule.

This gradient reflects the historical development of mathematics. Natural numbers have been recognized in all cultures since antiquity, while negative, rational, real, and complex numbers were progressively introduced as the operational needs of mathematics demanded them. The conceptual difficulty historically accompanying each extension — numeri ficti for negatives, the irrationality crisis in Greek mathematics, the long dispute over imaginary numbers — reflects, in retrospect, the increasing operational depth of these constructions.

VI.5. What Does Not Exist as an Object

In VR-Numbers, the following are not introduced as ontological objects:

Ordered pairs. Pairs appear nowhere in the construction as independent objects; pairing is replaced by syntactic juxtaposition ($a \ominus b$, $a \oslash b$, $a \oplus b{\cdot}i$).

Negative numbers. The integer -3 is a notation for the operational class $\{n \ominus (n{+}3) : n \in \mathbb{N}\}$, not an independent thing.

Fractions. The rational 1/2 is a notation for the class of equivalent division expressions, not a separate object.

Real numbers. A real number is an equivalence class of Cauchy sequences, which are themselves operational rules — finite specifications of infinite processes — not sets of ordered pairs.

Complex numbers. A complex number is a syntactic expression over two operational axes, not a two-dimensional point in a plane.

This minimalism is the philosophical position of VR-Numbers: only $\varnothing$ is. All else is doing.

VI.6. Consistency Relative to ZF

Each construction ($\mathbb{Z}$_VR, $\mathbb{Q}$_VR, $\mathbb{C}$_VR) is isomorphic to its standard counterpart; $\mathbb{R}$_VR is isomorphic to the computable subfield of the classical $\mathbb{R}$. The standard constructions of $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ from $\mathbb{N}$ are consistent relative to ZF, and the computable reals form a definable subfield of $\mathbb{R}$ within ZF. Since VR is consistent relative to ZF [Reznik 2026], and the constructions of this work yield structures definable within ZF, the full system of VR-Numbers is consistent relative to ZF.

Part VII. Open Questions

(1) Formalization in Type Theory.

The constructions of VR-Numbers — particularly the operational view of functions in the definition of $\mathbb{R}$, and the operational marker i in $\mathbb{C}$ — are naturally suited to constructive type theories such as Martin-Löf's. A formalization in Coq, Agda, or Lean would provide machine verification and explicit categorical semantics.

(2) Algebraic Closure Beyond $\mathbb{C}$.

By the Fundamental Theorem of Algebra, $\mathbb{C}$ is algebraically closed. This suggests that the two-axis construction of $\mathbb{C}$ via A1's duality is, in some sense, a natural endpoint of the operational gradient. The question whether further extensions (quaternions $\mathbb{H}$, octonions $\mathbb{O}$) admit analogous derivations from deeper logical structure remains open.

(3) The VR-Sets Program: Operational Set Theory.

The operational ontology of VR-Numbers — only $\varnothing$ is, all else is doing — invites a natural extension: an operational reconstruction of set theory itself. We outline this direction here as a program for future work, provisionally named VR-Sets.

The central observation is that the classical paradoxes of self-membership ($x \in x$) arise from a particular reading of the membership relation: that $x \in y$ means y physically contains x as a substructure. Under this reading, a set containing itself produces an irreducible structural cycle, and Russell's paradox follows. The axiom of foundation in ZFC is the explicit prohibition of such cycles.

In an operational ontology, the membership relation admits an alternative reading. The expression $x \in y$ can be understood as: x was used in the construction of y, or equivalently, y is a result of an operation referring to x. Under this reading, $x \in x$ is not paradoxical: it expresses that x is a result of an operation referring to its own name. This is analogous to recursive structures in programming: a recursive data structure does not physically contain itself, but it holds a reference to itself. Reference and containment are semantically distinct.

This distinction opens a possibility for unifying ZFC and ZFA — the classical foundational set theory and Aczel's anti-foundational set theory — as two operational modes of a single underlying system. In the ZFC-mode, references that create cycles are syntactically disallowed; the resulting system is provably equivalent to classical ZFC. In the ZFA-mode, cyclic references are admitted; the resulting system is equivalent to ZFA. The two systems, traditionally seen as incompatible alternatives, become restricted and unrestricted modes of the same operational set theory.

The technical formulation of VR-Sets requires introducing a layer of names alongside the operational objects. Sets are constructed through a sequence of operations: start(A) initiates a set under a name A; add(A, x) extends it with an element x; finish(A) seals the

construction. The expression A := {x, y, A} is then realized as start(A); add(A, x); add(A, y); add(A, A); finish(A). The final occurrence of A in add(A, A) is a reference to the name, not a physical inclusion of the object; the resulting set contains x, y, and a reference to itself, but no structural cycle in its construction. The two operational modes correspond to whether such name-cycles in references are admitted or forbidden.

Three open questions guide future work:

First, can the axioms of ZF be reformulated operationally over this construction, and proven equivalent in their classical force? This requires careful treatment of separation, replacement, and power set as operational schemes.

Second, does the operational reading dissolve Tarski's undefinability of truth, or merely relocate it? The duality of A1 ($F \to F$ and $F \to \top$) has structural similarity to self-reference; whether this duality has formal connections to truth predicates within VR-Sets is a natural avenue of investigation.

Third, can the axiom of choice find an operational interpretation that distinguishes its countable form (DC), available constructively, from its full form, classically necessary for non-constructive results such as Banach–Tarski? The operational ontology suggests that DC is natural (every operational rule can be carried out step by step) while full AC requires a separate justification.

(4) Computational Realization.

Each level of VR-Numbers admits a direct computational realization, making the system suitable as a foundation for exact-arithmetic computation systems.

Summary

VR-Numbers extends the formal system VR with operational superstructures yielding $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. The work advances three principal theses:

First, a minimal ontology: the single ontological primitive is the void $\varnothing$; all other objects, including the natural numbers, arise as operations of progressively greater depth performed over $\varnothing$. No ordered pairs are introduced as ontological objects.

Second, actual infinity is replaced by operational infinity: the axiom of induction A4 is read as an operational principle expressing universal applicability of an operation, not as a postulate of a completed infinite totality. The two notions agree on mathematical consequences but differ in ontological commitment.

Third, the construction of $\mathbb{C}$ is grounded in the duality of axiom A1: the two generating implications $F \to F$ and $F \to \top$ correspond to the real and imaginary axes, respectively. This provides an ontological justification for the two-dimensionality of $\mathbb{C}$ — a feature standardly postulated through pairs without deeper motivation. The joining rule $i^2 = \ominus 1$ binds the two axes into the algebraic structure of complex numbers.

The structures $\mathbb{Z}$_VR, $\mathbb{Q}$_VR, $\mathbb{C}$_VR are isomorphic to their standard counterparts; $\mathbb{R}$_VR is isomorphic to the field of computable real numbers (a countable subfield of classical $\mathbb{R}$), reflecting the constructive position of the work. The consistency of the system relative to ZF follows from the consistency of VR established in [Reznik 2026]. The philosophical position is minimal: only $\varnothing$ is — all else is doing.

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